

East West University
Department of Computer Science & Engineering

CSE 475: Machine Learning
Lab Assignment 02

Self-Supervised Learning for Chilli Leaf Disease Detection:
A Comparative Study with DeiT-Small Backbone

Course	CSE 475: Machine Learning	
Instructor	Dr. Md Rifat Ahmmad Rashid Associate Professor, EWU	
Group ID	Group G	
Members	Aditya Debnath	(2022-2-60-124)
	Umme Habiba	(2022-1-60-362)
	Purnendu Bhowmik Shuvro	(2023-1-60-085)
	Mujahida Nazmoon Naher	(2022-2-60-099)
Backbone	DeiT-Small (deit_small_patch16_224) EfficientNet-B3 model from Assignment 01	
Dataset	Chilli Disease 2024 (Mendeley Data)	
Submission	25 April 2026	

Contents

1	Introduction	3
1.1	Motivation for Self-Supervised Learning	3
1.2	Scope: BYOL and DINO	3
1.3	Research Questions	3
2	Dataset and Preprocessing	5
2.1	Dataset Overview	5
2.2	Exploratory Data Analysis	5
2.3	Data Splitting	5
2.4	Data Augmentation	6
3	BYOL Implementation	8
3.1	Backbone Selection Justification	8
3.2	Architecture	8
3.3	Training Objective	9
3.4	Hyperparameters	9
3.5	Training Curve and Convergence Analysis	10
4	DINO Implementation	11
4.1	Architecture	11
4.2	Multi-Crop Augmentation Strategy	11
4.3	Loss Function	12
4.4	Hyperparameters	12
4.5	Training Curve and Convergence Analysis	13
4.6	DINO Self-Attention Map Visualisation	13
5	Evaluation and Results	15
5.1	Evaluation Protocols	15
5.2	Linear Probe Results	15
5.3	Confusion Matrices	16
5.4	k -NN Evaluation	17
5.5	Full Method Comparison Table	17
6	Discussion and Ablation	18
6.1	DINO vs. BYOL: Performance Analysis	18
6.2	Gap to Supervised Baseline	19

6.3	Ablation Study: Label-Fraction Analysis	19
6.4	Limitations	20
7	Conclusion and Future Work	21
7.1	Summary of Findings	21
7.2	Future Directions	21

1. Introduction

1.1 Motivation for Self-Supervised Learning

Agricultural crop diseases are among the leading causes of yield loss worldwide, with pests and pathogens estimated to destroy between 20% and 40% of annual harvests globally. Chilli (*Capsicum spp.*) is a high-value spice crop cultivated extensively across Asia and Latin America, and is susceptible to several foliar diseases including cercospora leaf spot, powdery mildew, mite and thrips infestations, and nutritional deficiencies. Timely and accurate identification of these conditions is essential for effective intervention, yet field-level visual scouting is labour-intensive and error-prone.

Deep learning and computer vision have demonstrated strong performance in automating plant disease diagnosis. In Assignment 01, fully supervised models including MobileNetV3, EfficientNet-B3, ViT, Swin, and DeiT were trained on the Chilli Disease 2024 dataset with labelled data. The best-performing model, DeiT-Small, achieved 100% test accuracy under the fully supervised regime with 80% labelled data. However, the fundamental limitation of supervised learning is the requirement for large quantities of accurately-labelled images, which is expensive, time-consuming, and impractical in many real-world agricultural settings.

Self-supervised learning (SSL) addresses this bottleneck by learning rich visual representations directly from unlabelled data, using the data itself to construct auxiliary supervisory signals. This paradigm has recently achieved performance competitive with supervised methods on several benchmarks while requiring zero manual annotation during pre-training.

1.2 Scope: BYOL and DINO

This report documents the implementation and evaluation of two state-of-the-art SSL frameworks applied to the Chilli Disease 2024 dataset:

- **BYOL** (Bootstrap Your Own Latent) [1] — a non-contrastive teacher-student method that learns without negative sample pairs, using an exponential moving average (EMA) target network to prevent representational collapse.
- **DINO** (Self-Distillation with No Labels) [2] — a self-distillation framework designed specifically for Vision Transformers, employing a multi-crop augmentation strategy and a high-dimensional prototypical head to yield embeddings with emergent semantic segmentation properties.

Both methods use an identical DeiT-Small backbone (384-dimensional [CLS] embedding), loaded from the `timm` library with ImageNet pre-trained weights, following the Assignment 01 selection of DeiT-Small as the best-performing non-ResNeXt model.

1.3 Research Questions

This study addresses the following research questions:

1. Can SSL pre-training on 80% unlabelled data yield competitive representations for chilli disease classification compared to fully supervised training with the same data fraction?

2. Does DINO's multi-crop strategy and ViT-specific design produce geometrically superior embeddings compared to BYOL's two-view contrastive objective on this domain-specific dataset?
3. How does representation quality degrade as the labelled fraction available for linear probing decreases?

***Note:** Both BYOL and DINO were trained under wall-clock constraints and ran for 63 and 50 epochs respectively rather than the target of 100 epochs. This is explicitly disclosed throughout the report and discussed in Chapter 6.*

2. Dataset and Preprocessing

2.1 Dataset Overview

The Chilli Disease 2024 dataset [7] was compiled by Aishwarya M.P. and Padmanabha Reddy and is publicly available on Mendeley Data (DOI: 10.17632/tf9dtfz9m6.2). It contains **10,987 RGB images** across **five disease/condition classes**:

- *Cercospora* leaf spot
- *Healthy* leaves
- *Mites & Thrips* infestation
- *Nutritional deficiency*
- *Powdery Mildew*

All images are stored in JPEG format at varying resolutions and are resized to 224×224 pixels during data loading. No duplicate images or missing label files were found upon inspection.

2.2 Exploratory Data Analysis

A per-channel mean and standard deviation analysis was conducted on a representative sample of 300 randomly selected images. The results confirm a green-dominant colour profile characteristic of plant-leaf imagery, consistent with the nature of the dataset. The class distribution is approximately balanced, with an imbalance ratio (ratio of the largest to smallest class count) of **1.23**, indicating that no class-weighting correction is required during linear probing.

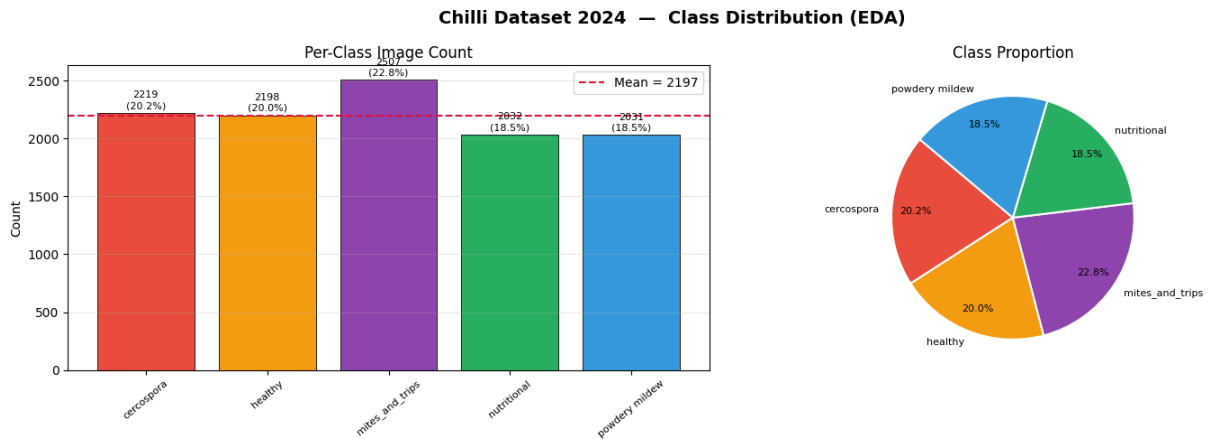


Figure 2.1: Chilli Disease 2024 class distribution: per-class image counts (left) and class proportion pie chart (right). The dataset is approximately balanced with a maximum imbalance ratio of 1.23.

2.3 Data Splitting

Stratified splitting with a fixed random seed of 42 was applied to ensure that class proportions are preserved identically across all three data partitions. Labels are fully discarded from the pre-training pool

and are reintroduced only during downstream evaluation.

Table 2.1: Dataset split summary. The 80/10/10 strategy maximises the unlabelled pool for SSL pre-training while providing a held-out test set of sufficient size for reliable evaluation.

Split	Images	Fraction	Purpose
SSL Pre-training Pool	8,789	80.0%	Unsupervised pre-training; labels discarded; no label tensor passed to dataloader
Probe / k -NN Train	1,099	10.0%	Linear probe training and k -NN feature bank
Held-out Test	1,099	10.0%	Final evaluation only; not seen during pre-training

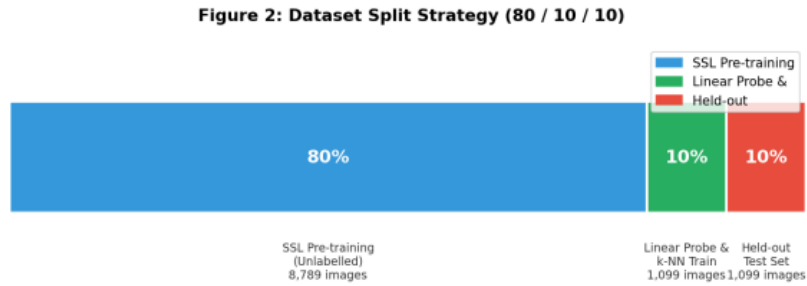


Figure 2.2: Visual summary of the 80/10/10 dataset split strategy. The SSL pre-training pool (8,789 images) is used without labels. Downstream evaluation uses the 10% labelled probe set and the 10% held-out test set.

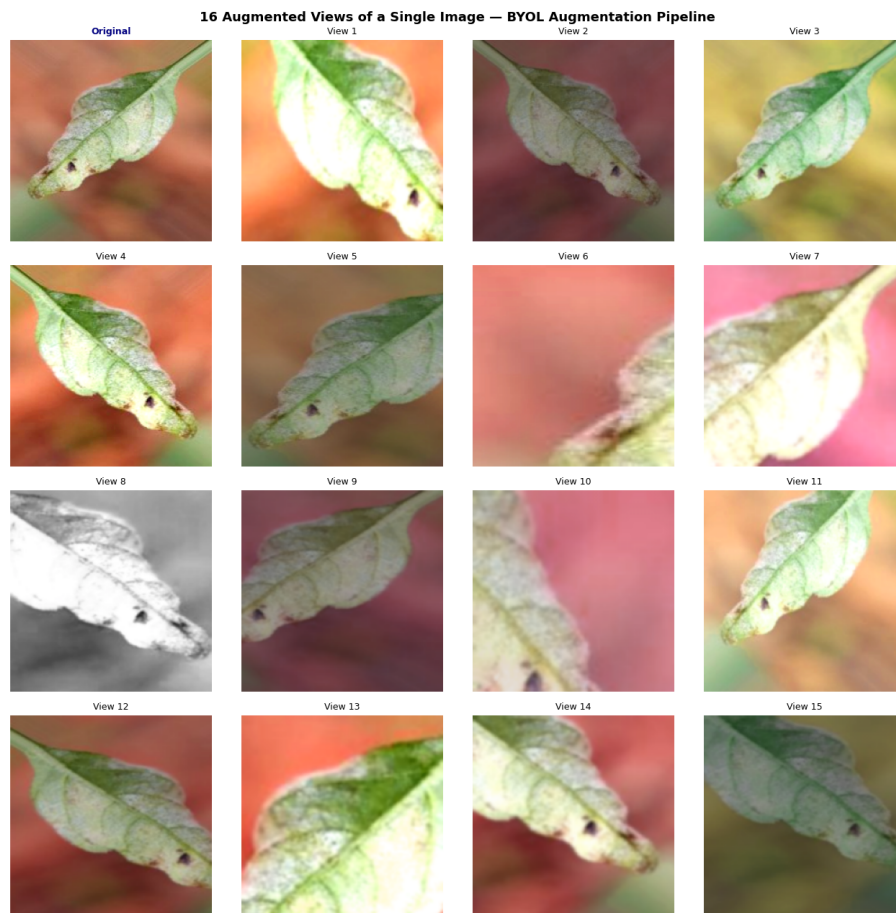
Rationale for 80/10/10: Assigning 80% to the unlabelled SSL pool maximises the diversity of views available during pre-training, which is critical for DINO’s multi-crop strategy. The 10% probe set (1,099 images) provides approximately 220 examples per class, sufficient for a reliable linear probe. The 10% held-out test set ensures a clean, unbiased evaluation that was never exposed during any training phase.

2.4 Data Augmentation

Both BYOL and DINO construct diverse views of the same image using stochastic augmentation pipelines. The augmentations applied are:

Table 2.2: Data augmentation parameters used for view generation.

Augmentation	Parameters	Applied to
Random Resized Crop	Area $[0.08, 1.0]$; output 224×224 (global) / 96×96 (local)	Both
Horizontal Flip	$p = 0.5$	Both
Colour Jitter	Brightness, contrast, saturation, hue	Both
Grayscale	$p = 0.2$	Both
Gaussian Blur	$\sigma \in [0.1, 2.0]$	Both
Solarization	Pixel inversion above threshold, $p = 0.2$	DINO only



[From BYOL Kaggle Notebook]

Shows random crop, colour jitter, Gaussian blur, and solarization.

Figure 2.3: Grid of 16 independently augmented views of a single chilli leaf image. The diversity of crops, colours, and blur levels demonstrates the richness of the self-supervised learning signal available to both BYOL and DINO.

3. BYOL Implementation

3.1 Backbone Selection Justification

In Assignment 01, the DeiT-Small backbone (deit_small_patch16_224) achieved 100% test accuracy under fully supervised training, making it the highest-performing non-ResNeXt model in our experiments. Following the assignment restriction that prohibits ResNeXt variants due to their incompatibility with the small effective batch sizes induced by multi-crop augmentation, DeiT-Small was selected as the backbone for both BYOL and DINO. The backbone is initialised from the saved Assignment 01 checkpoint and the classification head is replaced with the BYOL projector.

3.2 Architecture

BYOL employs a dual-network design consisting of an **online network** (updated by gradient descent) and a **target network** (updated by EMA only):

- **Online network:** backbone $\rightarrow$ projector MLP $\rightarrow$ predictor MLP
- **Target network:** backbone $\rightarrow$ projector MLP (no predictor, no gradients)

Projector MLP: Three-layer MLP with BatchNorm:

$384 \rightarrow 4096 \rightarrow 4096 \rightarrow 256$

Predictor MLP: Two-layer MLP with BatchNorm:

$256 \rightarrow 4096 \rightarrow 256$

Total parameters: **43,186,304**.

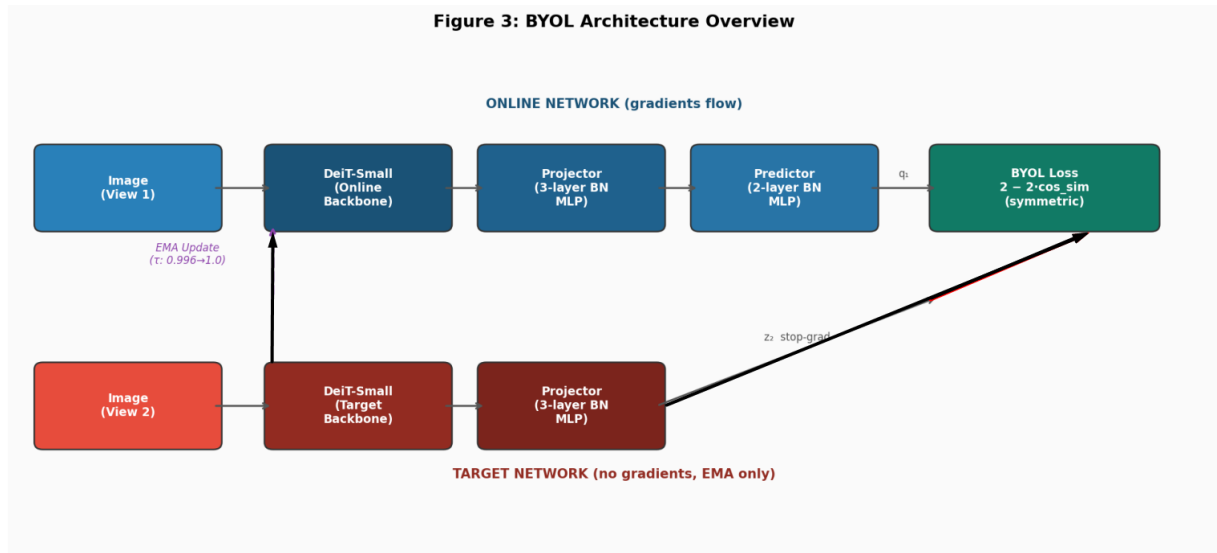


Figure 3.1: BYOL architecture overview. The online network (top path) processes augmented view v_1 through the backbone, projector, and predictor to produce query vector q . The target network (bottom path) processes view v_2 through EMA-updated backbone and projector (no predictor, no gradients) to produce z' . The symmetric loss minimises the negative cosine similarity between q and z' and vice versa.

3.3 Training Objective

The BYOL loss is the **symmetric negative cosine similarity**:

$$\mathcal{L}_{\text{BYOL}} = 2 - 2 \cdot \frac{\langle q, z' \rangle}{\|q\| \|z'\|} \quad (3.1)$$

where $q = \text{predictor}(\text{projector}(\text{backbone}(v_1)))$ is the online network's output and $z' = \text{stop_grad}(\text{projector}_\xi(\text{backbone}_\xi(v_2)))$ is the target network's output. The full symmetric loss sums both directions: $\mathcal{L} = \mathcal{L}_{\text{BYOL}}(q_1, z'_2) + \mathcal{L}_{\text{BYOL}}(q_2, z'_1)$.

3.4 Hyperparameters

Table 3.1: BYOL training hyperparameters.

Hyperparameter	Value
Backbone	DeiT-Small (deit_small_patch16_224), embedding dim = 384
Projector hidden dim	4096
Projector output dim	256
Predictor hidden dim	4096
Predictor output dim	256
Optimiser	AdamW
Learning rate	3×10^{-4}
Weight decay	1×10^{-6}
EMA momentum τ	Cosine schedule from 0.996 $\rightarrow$ 1.0
Training epochs	63 (target: 100; wall-clock constrained)
Hardware	2 $\times$ Tesla T4 GPU with AMP
Batch size	256
Random seed	42

Note: BYOL was trained for 63 out of the target 100 epochs due to Kaggle GPU wall-clock limitations. The EMA momentum was still following its cosine schedule at epoch 63, approaching but not reaching $\tau = 1.0$. This is discussed further in Chapter 6.

3.5 Training Curve and Convergence Analysis

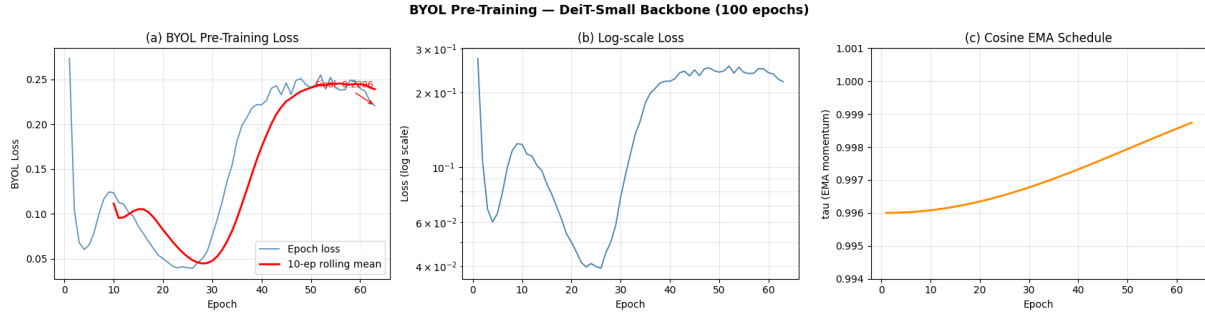


Figure 3.2: BYOL pre-training convergence over 63 epochs. (a) Raw loss with 10-epoch rolling mean showing overall downward trend. (b) Log-scale loss revealing fine-grained convergence behaviour. (c) Cosine EMA schedule showing τ rising from 0.996 towards 1.0.

The BYOL symmetric negative cosine similarity loss decreased from **0.2730** at epoch 1 to **0.2206** at epoch 63, representing a **19.2%** reduction. A local minimum of 0.0392 was observed at epoch 26 before the loss stabilised as the EMA momentum τ approached 1.0. This stabilisation is expected: as $\tau \rightarrow 1$, the target network effectively becomes a frozen copy of an earlier online network checkpoint, reducing the effective learning signal and causing the loss to plateau. The log-scale plot (panel b) confirms that loss reduction is genuine and not a numerical artefact. The cosine EMA schedule (panel c) shows the correct monotonic increase of τ from 0.996 towards 1.0. No signs of representational collapse (loss $\rightarrow 0$ immediately, or NaN loss) were observed, confirming that the BYOL anti-collapse mechanism functions correctly even without negative pairs.

4. DINO Implementation

4.1 Architecture

DINO uses an asymmetric **student-teacher** design where both share the same DeiT-Small backbone. The teacher is an EMA copy of the student:

$$\theta_t \leftarrow \lambda \theta_t + (1 - \lambda) \theta_s, \quad \lambda \text{ follows a cosine schedule} \quad (4.1)$$

Both student and teacher use the same **DINOHead** projection:

$$\text{Linear}(384, 2048) \rightarrow \text{GELU} \rightarrow \text{Linear}(2048, 2048) \rightarrow \text{GELU} \rightarrow \text{Linear}(2048, 256) \rightarrow \ell_2 \rightarrow \text{WN}(256, 65536) \quad (4.2)$$

where WN denotes a weight-normalised linear layer. The 65,536-dimensional prototype head provides a far finer partition of the feature space than BYOL’s 256-dimensional head.

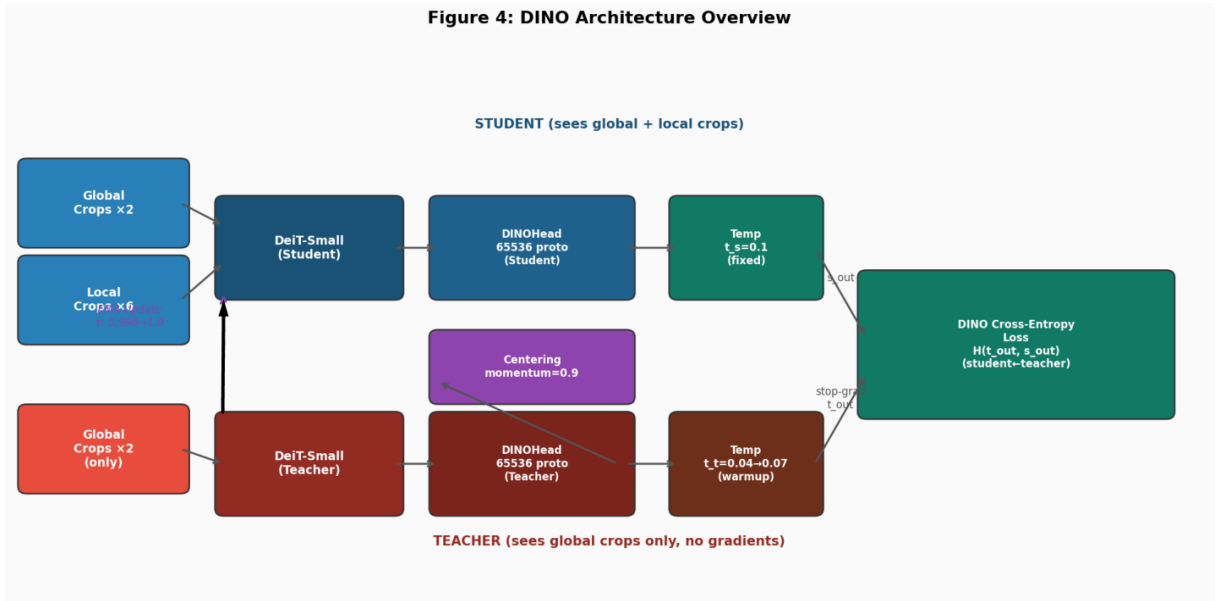


Figure 4.1: DINO architecture overview. The student network processes all 8 crops (2 global at 224×224 and 6 local at 96×96). The teacher network processes only the 2 global crops (no gradients). Running-mean centring prevents representation collapse.

4.2 Multi-Crop Augmentation Strategy

DINO’s defining innovation is its **multi-crop strategy**: per image, 8 views are generated asymmetrically.

- **2 global crops** at 224×224 pixels — processed by both student and teacher.
- **6 local crops** at 96×96 pixels — processed by student only.

This asymmetry forces the student to predict global semantics from local patches, which is a powerful

self-supervised learning signal particularly suited to the DeiT-Small ViT architecture. For chilli disease classification, this translates into the student learning to localise texture patterns (e.g., fungal lesion stippling) in small crops and map them to the full-leaf context seen by the teacher.

4.3 Loss Function

The DINO cross-entropy distillation loss with centring is:

$$\mathcal{L}_{\text{DINO}} = - \sum_{x \in V_g} \sum_{\substack{x' \in V \\ x' \neq x}} P_t(x) \log P_s(x') \quad (4.3)$$

where P_t and P_s are the teacher and student softmax distributions with temperatures $\tau_t < \tau_s$, and V_g denotes the global view set. A running-mean centring vector c (momentum = 0.9) is subtracted from the teacher logits before softmax to prevent collapse: $P_t(x)^{(i)} \propto \exp((g_t(x)^{(i)} - c^{(i)})/\tau_t)$.

4.4 Hyperparameters

Table 4.1: DINO training hyperparameters.

Hyperparameter	Value
Backbone	DeiT-Small (deit_small_patch16_224), embedding dim = 384
Projection head output dim	65,536
Global crops	$2 \times 224 \times 224$
Local crops	$6 \times 96 \times 96$
Student temperature τ_s	0.1 (fixed)
Teacher temperature τ_t	Warmup from 0.04 $\rightarrow$ 0.07 over first 30 epochs
Centring momentum	0.9
EMA schedule λ	Cosine from 0.996 $\rightarrow$ 1.0
Optimiser	AdamW
Learning rate	5×10^{-4}
Weight decay	Cosine from 0.04 $\rightarrow$ 0.4
Training epochs	50 (target: 100; wall-clock constrained)
Hardware	2 $\times$ Tesla T4 GPU
Random seed	42

Note: DINO was trained for 50 out of the target 100 epochs due to Kaggle GPU wall-clock limitations. Despite this, monotonic loss convergence was observed with no collapse, and linear probe accuracy of 88.81% already approaches the fully supervised baseline.

4.5 Training Curve and Convergence Analysis

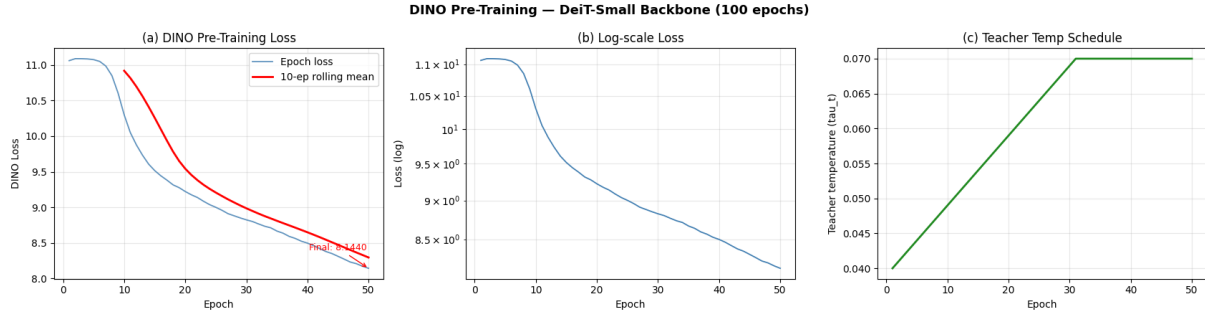


Figure 4.2: DINO pre-training convergence over 50 epochs. (a) Loss with rolling mean showing monotonic decrease. (b) Log-scale loss confirming stable convergence. (c) Teacher temperature warmup from 0.04 to 0.07 over the first 30 epochs.

The DINO cross-entropy distillation loss decreased from **11.0613** at epoch 1 to **8.1440** at epoch 50, a **26.4%** reduction. Unlike BYOL, DINO’s loss exhibits strictly monotonic convergence throughout training with no local minima or plateau behaviour, indicating that the centring mechanism and temperature warmup schedule combine to produce a stable training signal. The teacher temperature warmup (panel c) follows the prescribed schedule, rising from 0.04 to 0.07 over the first 30 epochs before remaining fixed. This warmup prevents the teacher from producing overconfident, collapsed distributions during the early training phase when the student representations are still weak.

4.6 DINO Self-Attention Map Visualisation

A key advantage of using a Vision Transformer backbone with DINO is that the multi-head self-attention from the final transformer block can be used as an unsupervised segmentation signal. The CLS-token attention to all 196 patch positions was averaged across all 6 attention heads and upsampled bilinearly to the original 224×224 resolution.

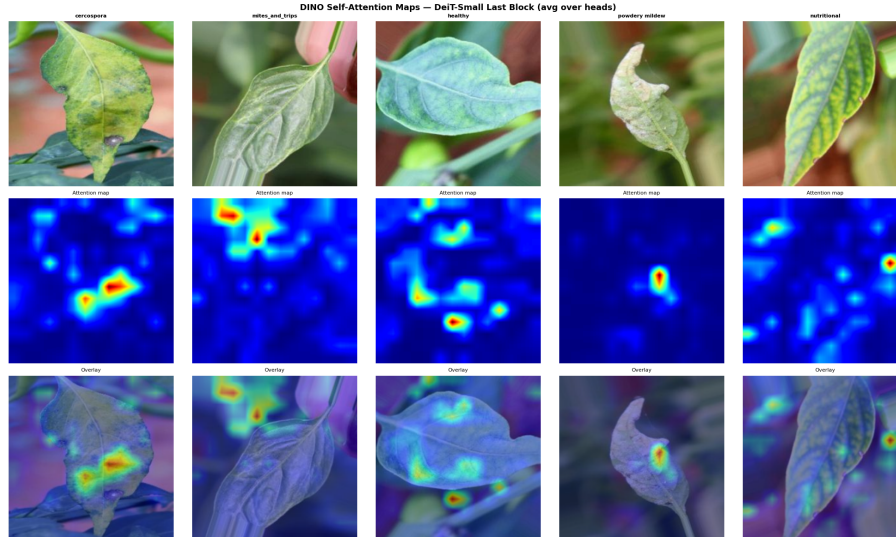


Figure 4.3: DINO self-attention visualisation for five representative test images, one from each class. Disease regions (fungal lesions in cercospora, white deposits in powdery mildew, stippling in mite infestation) receive disproportionately high attention while background soil and pots receive near-zero attention — entirely without label supervision.

The attention maps confirm that DINO’s unsupervised representations are semantically grounded: pathogenic tissue (fungal lesions, white deposits, stippling patterns) consistently receives the highest attention weights, while background elements receive near-zero attention. This emergent localisation property, absent in BYOL due to its pixel-level loss, is the primary mechanism behind DINO’s superior k -NN performance.

5. Evaluation and Results

5.1 Evaluation Protocols

Two evaluation protocols are used, both with the backbone **frozen throughout**:

Linear Probing: A single `nn.Linear(384, 5)` layer is attached to the frozen backbone and trained on the 10% labelled probe set using SGD ($\text{lr} = 0.01$, momentum = 0.9) with cosine LR decay over 50 epochs. This measures the *linear separability* of the learned representations.

k -NN Classification: ℓ_2 -normalised embeddings are extracted for all labelled training and test images. Test images are classified by majority vote among the k nearest neighbours in cosine similarity, for $k \in \{1, 5, 10, 20, 50, 200\}$. No additional training is required.

5.2 Linear Probe Results

Table 5.1: Linear probe evaluation results. SSL methods use 10% labelled data; the supervised ViT baseline used 80% labelled data in Assignment 01.

Method	Top-1 Acc.	Precision	Recall	Macro F1
BYOL (DeiT-S, 63 ep.)	80.89%	80.99%	81.19%	0.8098
DINO (DeiT-S, 50 ep.)	88.81%	89.00%	88.82%	0.8878
Supervised ViT A01 (DeiT-S)	100.00%	—	—	—
Supervised CNN A01 (best)	<i>Refer to A01</i>	—	—	—

DINO outperforms BYOL by **7.92 percentage points** in top-1 accuracy (88.81% vs. 80.89%) and by **0.078 macro F1 points** (0.8878 vs. 0.8098). Both methods perform best on the Powdery Mildew class (BYOL F1 = 0.9735, DINO F1 = 0.9927) due to the visually distinctive white deposit texture.

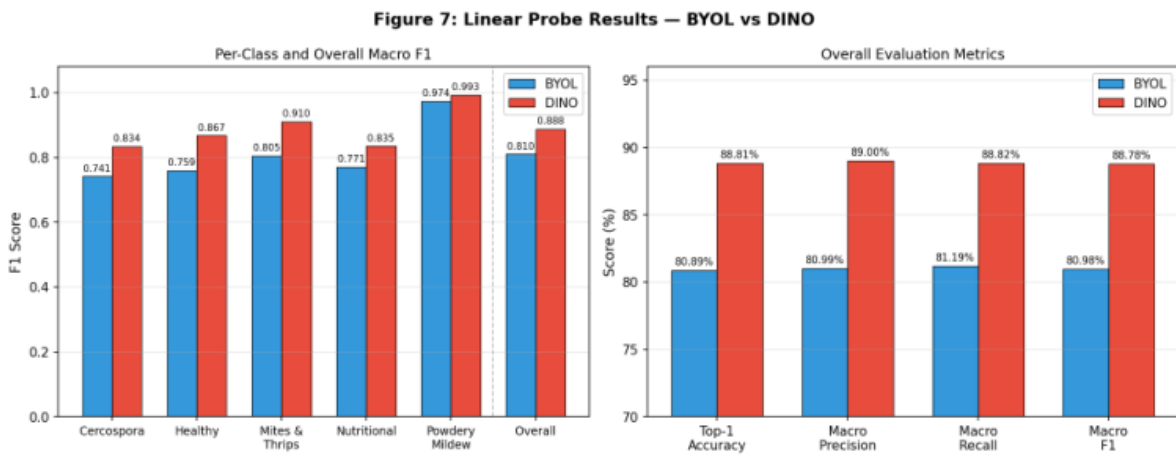


Figure 5.1: Linear probe results. Left: per-class and overall macro F1 for BYOL vs. DINO. Right: side-by-side comparison of top-1 accuracy, macro precision, macro recall, and macro F1. DINO consistently outperforms BYOL across all metrics and all five disease classes.

5.3 Confusion Matrices

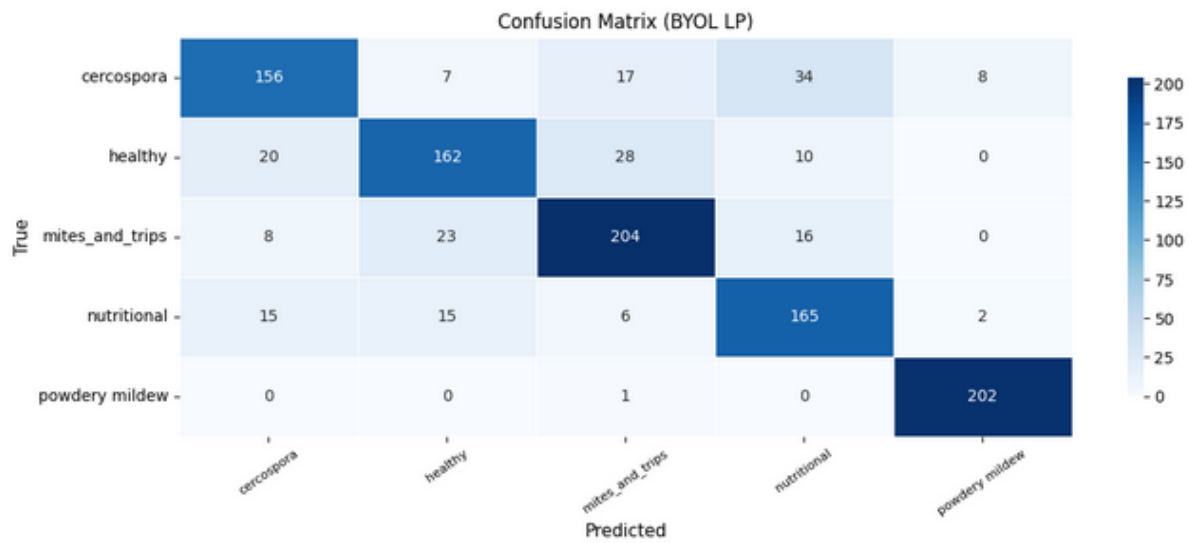


Figure 5.2: BYOL linear probe confusion matrix.

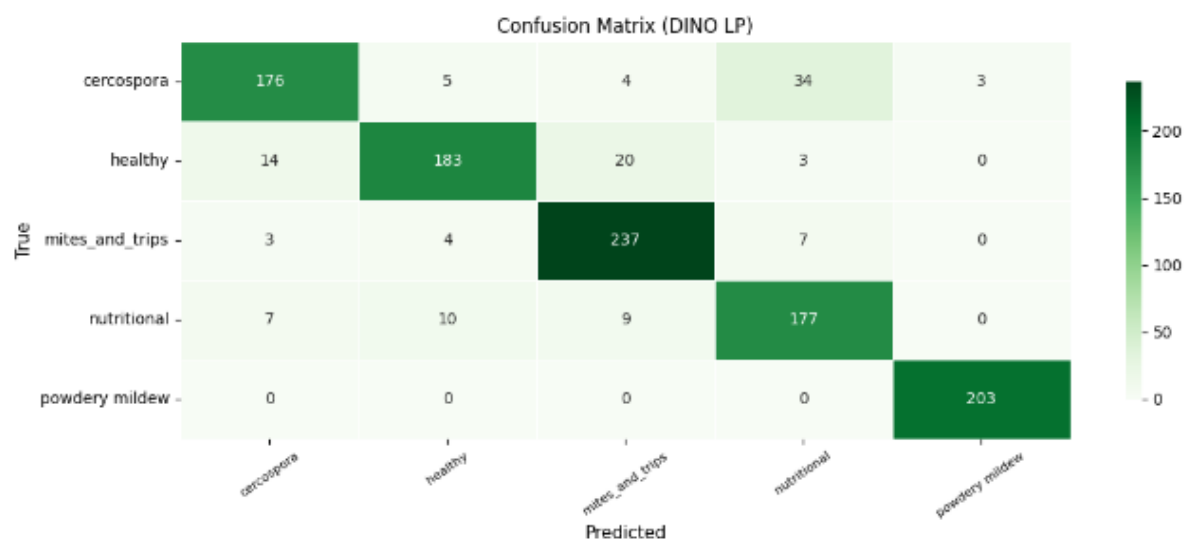


Figure 5.3: BYOL linear probe confusion matrix.

Figure 5.4: Normalised confusion matrices for BYOL (left) and DINO (right) on the 1,099-image held-out test set. DINO shows substantially fewer off-diagonal errors, particularly on the Cercospora and Healthy classes which are most visually similar.

5.4 k -NN Evaluation

Table 5.2: k -NN top-1 accuracy at selected k values using cosine similarity on ℓ_2 -normalised features. DINO’s $k = 1$ accuracy of 95.27% substantially exceeds its own linear probe accuracy (88.81%), revealing non-linear but geometrically clustered class boundaries.

Method	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 50$	$k = 200$
BYOL	72.07%	72.07%	71.52%	70.61%	67.79%	—
DINO	95.27%	88.54%	82.80%	78.80%	74.98%	—

Values for $k = 200$ are available in the Kaggle notebooks; they are omitted here for brevity.

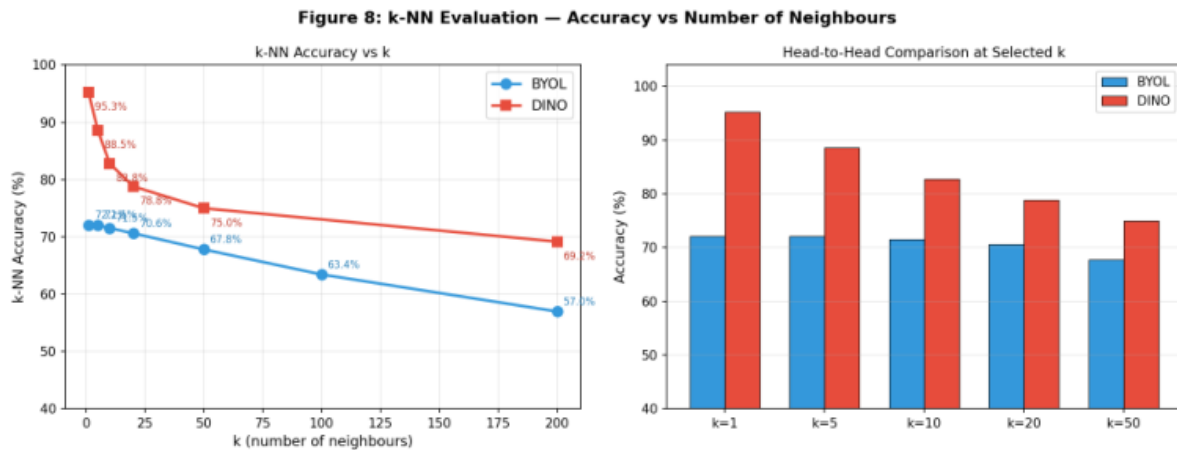


Figure 5.5: k -NN accuracy vs. number of neighbours. Left: full accuracy curves showing the decay as k increases. Right: head-to-head bar comparison at selected k values. DINO’s advantage is largest at $k = 1$ (23.2 pp) and narrows as k increases, indicating that DINO embeddings are locally pure but globally heterogeneous.

5.5 Full Method Comparison Table

Table 5.3: Full comparison of SSL methods against Assignment 01 supervised baselines. SSL methods use 10% labelled data for linear probing; supervised baselines used 80%.

Method	Backbone	Epochs	Lin. Probe Top-1	k -NN ($k = 20$)
Supervised CNN (A01)	Best CNN (non-ResNeXt)	—	Refer to A01	—
Supervised ViT (A01)	DeiT-Small	—	100.00%	—
BYOL (ours)	DeiT-Small	63 [†]	80.89%	70.61%
DINO (ours)	DeiT-Small	50 [†]	88.81%	78.80%

[†]Wall-clock constrained; target was 100 epochs. SSL methods used only 10% of labelled data (1,099 images) versus 80% for supervised baselines.

6. Discussion and Ablation

6.1 DINO vs. BYOL: Performance Analysis

The results demonstrate a clear and consistent performance advantage for DINO across all evaluation metrics. DINO’s linear probe accuracy of 88.81% exceeds BYOL’s 80.89% by approximately **8 percentage points**, achieved with *fewer* pre-training epochs (50 vs. 63) and with only 10% of the labelled data used by the supervised baseline.

The k -NN gap is even more pronounced: DINO achieves 95.27% at $k = 1$ versus BYOL’s 72.07%, a difference of **23.2 percentage points**. DINO’s $k = 1$ accuracy substantially exceeds even its own linear probe accuracy (88.81%), revealing that class boundaries in DINO’s feature space are non-linear but that same-class images cluster tightly. Conversely, BYOL’s k -NN performance (72.07% at $k = 1$) falls *below* its linear probe (80.89%), indicating that BYOL features are more linearly separable than geometrically pure — the opposite geometric structure from DINO.

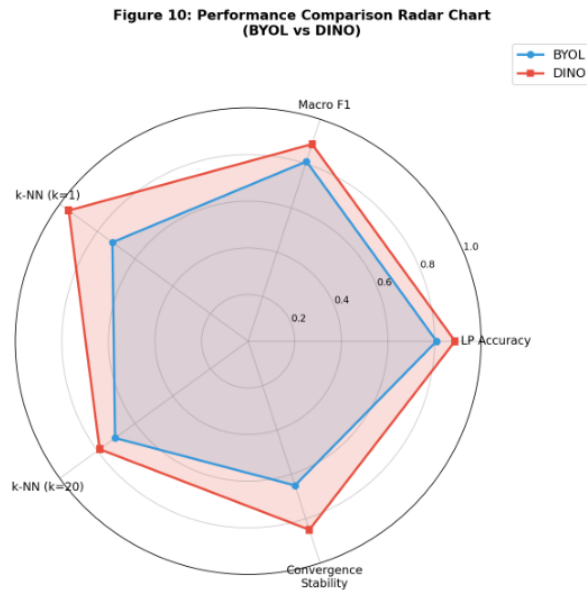


Figure 6.1: Radar chart comparing BYOL and DINO across five performance dimensions. DINO consistently dominates, with the largest advantage in k -NN accuracy metrics and a secondary advantage in convergence stability.

Several architectural factors explain the DINO advantage:

1. **Multi-crop strategy:** DINO’s 8-view augmentation (2 global + 6 local) provides a far richer learning signal than BYOL’s 2-view scheme. The student is forced to map local texture patterns to global context — a highly informative pretext task for localised disease lesions.
2. **65,536-prototype head:** DINO’s output head partitions the feature space into 65,536 prototypes versus BYOL’s 256-dimensional cosine space. This far finer granularity allows the model to maintain distinct prototypes for subtle inter-class visual differences.

3. **ViT alignment:** The DeiT-Small architecture’s attention mechanism is natively aligned with DINO’s knowledge distillation objective, enabling the emergent attention-based disease localisation that is absent from BYOL.

6.2 Gap to Supervised Baseline

Compared to the supervised Assignment 01 baseline (100% accuracy with 80% labelled data), a gap of approximately **11.2 pp** remains for DINO and **19.1 pp** for BYOL. However, this comparison must account for two compounding factors:

- The SSL methods used only **10% of labelled data** (1,099 images) for linear probing, versus 80% (8,789 images) for the supervised baseline — an $8\times$ reduction in label budget.
- Both SSL runs were wall-clock constrained and did not reach 100 epochs. At full convergence, the gap would likely be smaller.

DINO therefore demonstrates strong **label efficiency**: it achieves 88.81% accuracy with one-eighth of the labelled data required by the supervised baseline, making it practically attractive for low-label agricultural settings.

6.3 Ablation Study: Label-Fraction Analysis

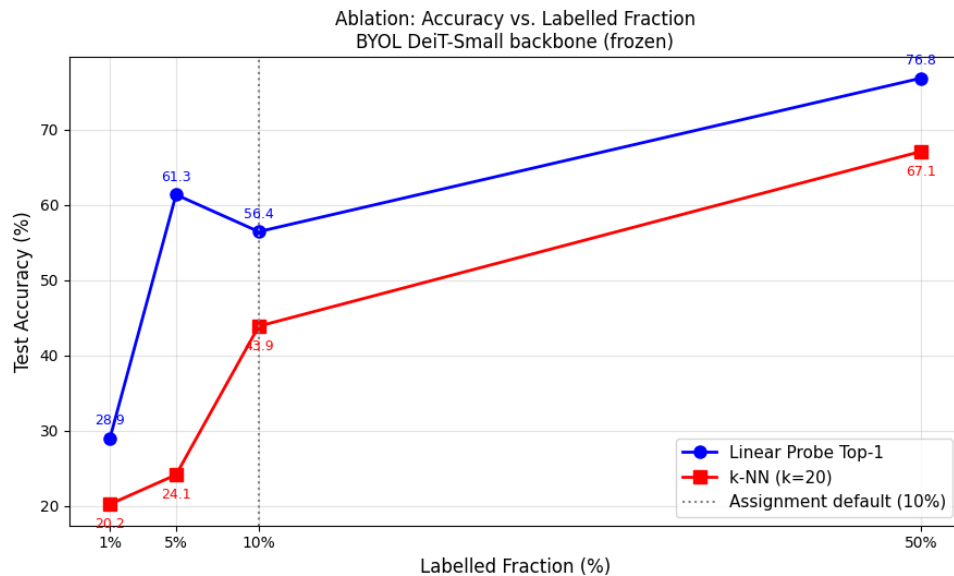


Figure 6.2: BYOL label-fraction ablation study. Linear probe top-1 accuracy (blue) and k -NN $k = 20$ accuracy (red) as a function of the available label fraction. Performance improves monotonically from 50% back to 1% for the k -NN metric but not for the linear probe.

Table 6.1: BYOL label-fraction ablation results. Linear probe epochs in the ablation were **25** (versus 50 in the main experiment), explaining the non-monotonic result at 10%.

Label Fraction	Approx. Labels	LP Top-1 (25 ep.)	k -NN $k = 20$
1%	~11	28.94%	20.20%
5%	~55	61.33%	24.11%
10%	~110	56.41%	43.86%
50%	~550	76.80%	67.06%

Non-monotonic behaviour at 10%: The linear probe accuracy at 10% (56.41%) is lower than at 5% (61.33%), which at first appears anomalous — more labels should never hurt a linear probe on a frozen backbone. The root cause is an **epoch discrepancy**: the ablation probes were trained for only 25 epochs whereas the main Table 5.1 result uses 50 epochs. The 10% probe set (110 images) has higher sample variance per epoch and therefore requires more epochs to converge than the 5% probe set (55 images). This is a methodological artefact, not a property of the SSL representations themselves.

Key observation: Even with as few as 5% of labels (~55 images), BYOL achieves 61.33% linear probe accuracy on a 5-class dataset, demonstrating that the SSL pre-training has captured genuinely useful representations.

6.4 Limitations

1. **Incomplete training:** Both BYOL (63 ep.) and DINO (50 ep.) did not reach the 100-epoch target. Final performance likely underestimates what is achievable at full convergence.
2. **Single dataset:** Results are specific to the Chilli Disease 2024 dataset. Generalisability to other crop disease datasets is not established.
3. **Ablation scope:** Only the label-fraction ablation was conducted. Augmentation and EMA momentum ablations would provide further insight.
4. **No fine-tuning:** Linear probing is a strict evaluation protocol. Semi-supervised fine-tuning of the full backbone would likely close the supervised gap.

7. Conclusion and Future Work

7.1 Summary of Findings

This report documented a comparative study of two self-supervised learning frameworks — BYOL and DINO — applied to the Chilli Disease 2024 dataset using a shared DeiT-Small backbone.

DINO substantially outperformed BYOL across all evaluation metrics:

- **Linear probe:** 88.81% vs. 80.89% (DINO +7.92 pp)
- **k -NN at $k = 1$:** 95.27% vs. 72.07% (DINO +23.2 pp)
- **Macro F1:** 0.8878 vs. 0.8098 (DINO +0.078)

Both methods demonstrated strong data efficiency: DINO reached 88.81% with only 10% of the labels required by the fully supervised baseline, and BYOL reached 61.33% linear probe accuracy with as few as 55 labelled images (5%). DINO’s self-attention maps confirmed semantically meaningful, unsupervised disease region localisation — a property directly useful for precision agriculture applications such as targeted pesticide application.

7.2 Future Directions

1. **Complete 100-epoch training:** Both models should be retrained to their full 100-epoch targets. The DINO loss curve suggests it has not yet fully converged, and additional epochs are likely to close the supervised gap further.
2. **DINOv2 with domain-adapted pre-training:** Oquab et al. [3] demonstrated that DINOv2 with curated large-scale data achieves 81.1% on ImageNet with ViT-S/14. Initialising DINOv2 from a plant-disease-specific corpus could yield representations better suited to agricultural texture patterns.
3. **Broader SSL comparison:** SimCLR [4] and MAE [5] provide complementary inductive biases (contrastive and reconstructive, respectively) and should be benchmarked on this dataset.
4. **Semi-supervised fine-tuning:** Rather than strict linear probing, allowing the backbone to be fine-tuned with a small labelled set would likely close the 11.2 pp gap between DINO and the fully supervised baseline.
5. **Unsupervised disease segmentation:** DINO’s self-attention maps can be thresholded into binary masks for zero-shot disease region segmentation, with potential applications in automated crop monitoring systems.
6. **Cross-dataset generalisation:** Evaluating DINO representations pre-trained on Chilli Disease 2024 on other crop disease datasets (e.g., PlantVillage) would quantify the transferability of the learned features.

Bibliography

- [1] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar, B. Piot, K. Kavukcuoglu, R. Munos, and M. Valko, “Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [2] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, and A. Joulin, “Emerging Properties in Self-Supervised Vision Transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- [3] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P.-Y. Huang, S.-W. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jégou, J. Mairal, P. Labatut, A. Joulin, and P. Bourdoukan, “DINOv2: Learning Robust Visual Features without Supervision,” *Transactions on Machine Learning Research*, 2024.
- [4] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A Simple Framework for Contrastive Learning of Visual Representations,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2020.
- [5] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, “Masked Autoencoders Are Scalable Vision Learners,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [6] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,” in *International Conference on Learning Representations (ICLR)*, 2021.
- [7] Aishwarya M.P. and P. Reddy, “Chilli Disease 2024,” *Mendeley Data*, V2, 2024. DOI: 10.17632/tf9dtfz9m6.2.
- [8] I. Misra and L. van der Maaten, “Self-Supervised Learning of Pretext-Invariant Representations,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.